

**B.Sc. (HONS.) IN CSE FIRST YEAR SECOND SEMESTER
EXAMINATION, 2021**

STATISTICS AND PROBABILITY

[According to the New Syllabus]

Subject Code : 510227

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

Marks

1. (a) Define Statistics. Discuss the importance and uses of Statistics. 6
- (b) Define data. What are the various types of data available for statistical analysis? 6
- (c) What is frequency distribution? The obtained marks of 30 candidates in an examination are given below : 8

34	36	31	46	76	86	42	44
32	46	40	54	66	56	50	42
33	80	77	81	46	40	60	63
64	76	56	57	57	70		

- (i) Construct a frequency distribution table by using suitable class interval.
- (ii) Construct a stem and leaf diagram for this data.
2. (a) Distinguish between : 5
- (i) Qualitative data and quantitative data.
- (ii) Discrete variable and continuous variable.
- (b) Describe various measures of central tendency. 5
- (c) For two non-zero positive numbers. Prove that, $A.M. \times H.M. = (G.M.)^2$. 5
- (d) Mean of 540 observations is found to be 60. If at the time of calculation two terms are wrongly taken as 77 and 67 instead of 47 and 57. Find the correct mean. 5
3. (a) What is questionnaire? What precautions should be taken in drafting a questionnaire? 8
- (b) Which measure is suitable for measures of dispersion and why? 5
- (c) Define classification. Describe the different types of classification. 7

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4. (a) Define measures of dispersion. Describe the characteristics of an ideal measure of dispersion. 4
- (b) Describe the merits and demerits of mean deviation and standard deviation. 6
- (c) Differentiate between standard deviation and coefficient of variation. 5
- (d) The average weekly wages and standard deviations of two factories are as follows : 5

Factory	Average weekly wages	Standard deviation	No. of workers
A	460	50	100
B	490	40	80

- (i) Which factory pays larger amount for weekly wages?
- (ii) Which factory shows larger variability in the distribution of wages?

5. (a) What is Kurtosis? Explain the different kinds of Kurtosis with the help of graph. 5
- (b) Following are the heights and weights of 8 students of B.Sc. (Hons.) class : 5

Height (inch)	62	72	68	64	70	66	64	72
Weight (kg)	50	65	64	56	62	63	60	64

Draw a scatter diagram and indicate whether the correlation is positive or negative.

- (c) Karl Pearson's coefficient of correlation between two variates x and y is 0.75, standard deviation of x is 3 and their covariance is 8.2. Find the variance of y . 5
- (d) The worker's age are given below : 5
- Age (year) : 25, 27, 30, 31, 33.
- Determine the first four central moments.

6. (a) Define multiple and partial correlations. 5
- (b) Discuss the different nature of correlation with the help of scatter diagram. 5
- (c) Calculate Karl Pearson's coefficient of correlation from the following data : 5

x	10	12	21	4	7
y	8	7	17	2	6

- (d) If the standard deviation of variable x is 6 and the two regression equations are $12x - 15y + 99 = 0$ and $60x - 27y = 321$. 5

Find the average value of x and y .